Literature Survey: Bayesian Deep Learning for SIC Failure Prediction and Reliability Aware NOMA Scheduling

Machine Learning 2nd Review · Course 25SC2107E · Department of ECE · A.Y. 2026–27

1. Scope and Method

This survey reviews the literature behind the project's three components: power-domain NOMA with successive interference cancellation (SIC), deep learning applied to NOMA receivers and user pairing, and Monte Carlo (MC) dropout as an uncertainty estimate for neural predictors. It also covers calibration and evaluation of probabilistic classifiers, since the project scores predictions with ROC-AUC and the Brier score.

The review draws on the project report and slides, a limited web search for recent NOMA and deep-learning papers, and well-known foundational references. Several sources were checked only from abstracts or search snippets. Items marked (verify) should be checked against the original paper before the survey is submitted.

2. NOMA and SIC Fundamentals

Non-orthogonal multiple access is widely treated as a route to higher spectral efficiency in 5G and beyond. Dai et al. give an early overview of NOMA solutions, challenges and research directions for 5G, placing power-domain NOMA with SIC at the centre of the discussion. Ding, Yang, Fan and Poor analyse NOMA performance when users are randomly deployed, showing that the gain depends strongly on how users are grouped and how much power separation exists between them. Ding et al.'s later survey in the IEEE Journal on Selected Areas in Communications extends this to open problems including user pairing, power allocation and receiver design. Saito et al. introduced NOMA as a candidate radio access scheme for cellular systems, which is the earliest framing of the strong-user/weak-user structure used in this project.

The core mechanism is consistent across these works. The receiver of the stronger user decodes the weaker user's higher-power signal, reconstructs it, subtracts it, and then decodes its own signal. Each step depends on the previous one, so an error in the first decision propagates. This error propagation is the failure mode the project models.

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3. Imperfect SIC

The perfect-SIC assumption is common in analytical NOMA papers, but practical receivers face channel-estimation error, residual interference after cancellation, and decoding errors. Much of the imperfect-SIC literature models residual interference as a fraction of the cancelled signal power and then derives outage or rate expressions.

The project takes a different route. It does not derive closed-form expressions. It generates synthetic SIC attempts with a latent failure probability built from link quality and non-ideal factors (channel-estimation error, power gap, mobility). This is a reasonable choice for a learning-focused prototype, but it means the classifier learns the simulator's own structure. The survey should state this clearly, because the ROC-AUC of 0.928 reflects how well the classifier recovers a rule the project itself wrote.

4. Deep Learning for NOMA Receivers

A growing body of work replaces or augments the SIC chain with learned models. Elfikky et al. (verify venue: Sensors) survey deep-learning-based NOMA, covering SIC, channel-state information, power and resource allocation, user fairness and transceiver design, and list open challenges. Panda and Singh (2022) analyse deep-learning NOMA receivers that detect multiple users without explicit channel knowledge, comparing several sequence models against conventional SIC on OFDM symbols.

A 2021 study on deep-learning SIC in NOMA (Ajou University repository record, verify authors and venue) uses a convolutional network and examines imperfect SIC explicitly. It is the closest prior work to this project's problem setting. DL-NOMA-CancelNet (Digital Engineering and Digital Society, verify) combines CNN and BiLSTM layers to reconstruct user signals directly, arguing that this avoids error propagation in conventional SIC.

These works share a common aim: make the receiver itself more robust. The present project is different in kind. It does not replace SIC. It predicts when SIC is likely to fail and uses that prediction upstream, at the scheduling stage. This placement is less common in the literature and is the project's main conceptual contribution.

5. User Pairing and Scheduling

User pairing determines which users share a resource and therefore how hard SIC is. Classical pairing rules sort users by channel gain and pair the strongest with the weakest, which is the conventional baseline used in this project.

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Recent work replaces this rule with learned or optimised pairing. A pointer-network deep reinforcement learning scheme (Chinese-language journal, verify) treats pairing as a combinatorial problem under imperfect SIC and jointly optimises transmit power to maximise reachable rate. An arXiv study on joint user pairing and association in multicell NOMA (arXiv:2004.07395) uses a pointer-network approach and reports roughly 30 percent gains over random pairing. A deep-learning framework for user clustering and pairing in UAV-assisted NOMA (recent conference submission, verify) uses a multilayer perceptron to learn pairing patterns with much lower complexity than exhaustive search.

The key distinction is what the learned component optimises. The reviewed pairing papers learn to maximise rate or sum-throughput, usually with reinforcement learning or supervised imitation of an optimiser. The project instead learns a failure probability and an uncertainty, then solves a minimum-cost assignment over those quantities. The cost (predicted failure probability plus 0.50 times standard deviation) is a risk-aware objective rather than a rate objective. The survey should present it as an alternative, not as an improvement on existing pairing algorithms, since no comparison with those algorithms has been run.

6. Uncertainty Estimation with Dropout

Gal and Ghahramani (2016) showed that a network trained with dropout can be interpreted as an approximate Bayesian model. Keeping dropout active at test time and averaging several stochastic passes yields a predictive mean and a spread. This is the foundation of the project's MC-dropout component.

Kendall and Gal (2017) separate aleatoric from epistemic uncertainty and argue that both are needed in Bayesian deep learning for vision. Lakshminarayanan, Pritzel and Blundell (2017) propose deep ensembles as a competitive and simpler alternative. Blundell et al. (2015) describe Bayes by Backprop, a variational method that treats weights as distributions. Labach, Salehinejad and Valaee (2019) survey dropout variants and their uses, including uncertainty.

A caution runs through this literature and applies directly here. The standard deviation across dropout passes approximates epistemic spread under one specific variational family. It is not a calibrated posterior variance, and it can be small even when the model is wrong. The project report already acknowledges this. The survey should emphasise it, because the scheduler gives uncertainty a weight of 0.50 with no sensitivity analysis.

7. Uncertainty in Wireless Deep Learning

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MC dropout has begun to appear in wireless applications. Catak (2023, IEEE MECO, verify) uses MC dropout with pilot-based channel estimation as a prior, claiming uncertainty-aware and more trustworthy 5G channel estimates. A 2021 IEEE Signal Processing Letters paper (vol. 29, pp. 269–273) applies MC dropout to CNN-based mmWave MIMO localization and shows that the resulting uncertainty regions depend on received power. This is a useful precedent for linking uncertainty to link quality. Ma and Kaewell (arXiv:2001.11963) apply a fast MC-dropout variant to radio-transmitter classification and report better uncertainty than simple ensemble averaging, while noting that the method can flag transmitters not seen in training.

These papers establish that MC dropout is used in wireless settings. Most evaluate uncertainty qualitatively or through localization error bands, and few check whether predicted probabilities are calibrated against observed outcomes. That is the gap this project inherits.

8. Calibration and Evaluation of Probabilistic Predictions

Accuracy and ROC-AUC measure ranking and thresholded decisions, but not whether a predicted 0.2 corresponds to a 20 percent failure rate. The Brier score (Brier, 1950) measures mean squared probability error and is the right companion metric here, since the scheduler consumes probabilities.

Niculescu-Mizil and Caruana (2005) show that many classifiers are miscalibrated in characteristic ways. Guo et al. (2017) show that modern neural networks are often overconfident, with temperature scaling as a simple remedy. A reliability diagram and expected calibration error are the standard next step for this project. The report lists them as future work, which is appropriate. Still, uncalibrated probabilities would weaken the pairing cost even if ROC-AUC is high.

9. Gap Analysis

Area	What the literature covers	What this project adds	Remaining gap
NOMA/SIC theory	Mature; analytical outage and rate under perfect and imperfect SIC	Simplified synthetic link model	No analytical benchmark for the simulator
DL for NOMA receivers	End-to-end detectors replacing SIC	Predicts SIC failure instead of replacing SIC	Not compared against DL receivers

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Area	What the literature covers	What this project adds	Remaining gap
User pairing	Rate-optimised pairing via RL or pointer networks	Risk-aware pairing via predicted failure and uncertainty	No comparison with learned or optimal pairing baselines
MC dropout in wireless	Applied to channel estimation and localization	Applied to SIC failure risk for scheduling	Uncertainty not validated against outcomes
Calibration	Well studied in general ML	Reports Brier score	No reliability diagram or ECE yet

10. Implications for the Project

The literature supports the project's framing on three points. SIC failure is a real practical concern and a sensible prediction target. Pairing is an established lever on SIC difficulty, so a scheduler that uses receiver-side risk is well motivated. MC dropout is a recognised, low-cost approximation for uncertainty in wireless deep learning.

Three points need strengthening before final submission. The first is baselines: the scheduler should be compared at least with random pairing and with a rate-optimised or risk-only pairing, as the report already suggests. The second is calibration: a reliability diagram and ECE would show whether the mean probability can be used directly in the cost. The third is replication: the 40-pair result comes from one seed and one user set, so mean and spread over several seeds should be reported.

A related internal inconsistency also needs resolving. The slides cite 60,000 synthetic attempts, while the code and report use 7,000. Whichever value is correct, the survey and presentation must describe the same experiment.

11. Suggested Reading Priorities

For a reader new to the area, the best starting points are Dai et al. (2015) for NOMA context, Ding et al. (2017) for open problems, Gal and Ghahramani (2016) for the Bayesian reading of dropout, Guo et al. (2017) for calibration, and the Elfikky et al. survey for deep learning in NOMA. The imperfect-SIC pairing paper and the 2021 deep-learning SIC study are the closest prior work and should be read in full.

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[22] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," ICML, 2017.

Note on the Source Documents

The reference [4] in the project report (Dietterich, 2000, on ensembles of decision trees) is only loosely related to the implemented method and may be dropped or justified explicitly. The project report's reference list has no imperfect-SIC or DL-NOMA papers. Adding items [5] to [11] above would strengthen the background chapter.

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